Aurick Anwar

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Education

McMaster University Hamilton, Ontario
Bachelor of Engineering: Engineering Physics *Expected 2029*

Work Experience

Software Engineering Intern Toronto, Ontario
HermesAI *May 2026 – Present*

- Developed **React.js** features for an AI legal-tech platform supporting **1,000+** page case files.
- Built scalable legal document dashboards, cutting manual processing time by **50%** for Canadian law firms.
- Optimized UI rendering and state management, boosting platform responsiveness by **30%**.

Founding Engineer Toronto, Ontario
Magnified Systems *February 2026 - Present*

- Building a helmet-mounted device to detect and quantify **impact severity in real time.**
- Trained a **PyTorch** regression model predicting impact severity (1–100 scale) at **96.4%** accuracy.
- Developed a severity-score algorithm in **C++** using linear and angular acceleration data.
- Designed the 3D modelled device enclosure in **SolidWorks**, producing a **PETG** fabricated model.

Manufacturing Assistant Toronto, Ontario
The STEAM Project June 2025 – August 2025

- Co-developed a Battle Bots curriculum teaching **SolidWorks** & **Arduino** to **50+** students, raising project completion by ~**40%**.
- Coordinated packaging and delivery of **100+** project kits across multiple locations.

Media & Operations Lead Toronto, Ontario
Orbitview December 2023 – June 2025

- Led **15+** developers building scalable media platforms and AI webinars with **200+ attendees** and **5× ROI**.
- Directed AI-focused events generating **$1,500+** in revenue through startup collaborations.

Personal Projects

↗ **Autonomous Self Driving Vehicle with CARLA** [*CARLA, YOLOv11, PyTorch, OpenCV*]

- Built an end-to-end autonomous driving pipeline in CARLA integrating **YOLO** object detection, achieving real-time simulation at ~**20–30 FPS**.
- Generated a **25k+** sample dataset with **5+ features** (steer, brake, speed, acceleration, throttle) to train a **PyTorch** model.
- Achieved ~**88.5%** directional prediction accuracy using Cross Entropy Loss for precise navigation.

↗ **AI Smart Computer w/ Hand Gesture Control** [*MediaPipe, PyAutoGUI, OpenCV*]

- Implemented **6+ gesture mappings** with 21-point hand tracking for click, cursor, mute, and volume control.
- Reduced unintended input by ~**60%** via gesture filtering and cooldown logic, achieving **<100 ms** latency.
- Tested with **10+** users with dexterity issues, improving task speed by ~**45%** and cutting mis-gesture errors by **70%**.

↗ **Google Home Replica** [*Python, Google Cloud, Text-to-speech*]

- Built a Google Home/Alexa replica using **Google Cloud APIs** and **Text-to-Speech**, processing queries in **<3s**.
- Implemented **4+ command types** (math calculations, weather) enabling real-time voice interaction.

Technical Skills

Programming: Python, C++, React.JS, CSS, HTML, JavaScript, ROS2
Libraries: PyTorch, OpenCV, MediaPipe, YOLO, Numpy, Pandas, scikit-learn
Simulation: RVIZ, SLAM, Gazebo, Nvidia Isaac Sim
3D Design: AutoCAD, Fusion360, Inventor, SolidWorks
Electronics and Microcontrollers: Arduino, Raspberry PI, ESP32
Tools & OS: Linux, Git, GitHub